

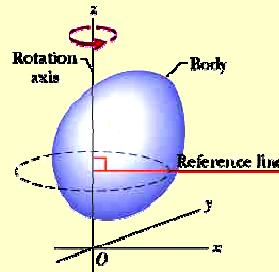


Rotation

Rigid Bodies
Rotation variables
Constant angular acceleration
Rotational KE
Rotational Inertia

Rotational Variables

- Rotation of a **rigid body**
- About a fixed **rotation axis**.
- **Rigid Body** – an object that can rotate with all its parts locked together and without any change in its shape.
- **Fixed axis** – Rotation occurs about an axis that does not move.
- **Every point** of the body moves in a **circle**
- whose **center lies on the rotation axis**,
- and moves through the **same angle** during a **time interval**.



Rotational Variables

Angular position:

$$\theta = \frac{s}{r}$$

Angular displacement:

$$\Delta\theta = \theta_2 - \theta_1$$

Average angular velocity:

$$\bar{\omega} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t}$$

Average angular velocity:

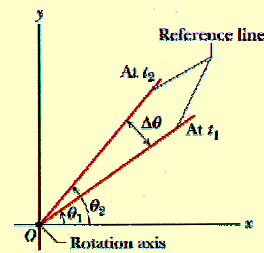
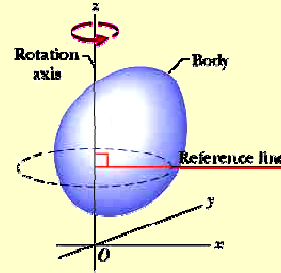
$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

Average angular acceleration:

$$\bar{\alpha} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t}$$

Instantaneous angular acceleration:

$$\alpha = \frac{d\omega}{dt}$$



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Rotational Variables

- Reference line $\perp$ to rotation axis
- **Rule:** An angular displacement in the counter-clockwise direction is positive, and one in the clockwise direction is negative.
- θ is the angle in radians – we **do not reset** θ to 0 after each rotation.

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Problem 2

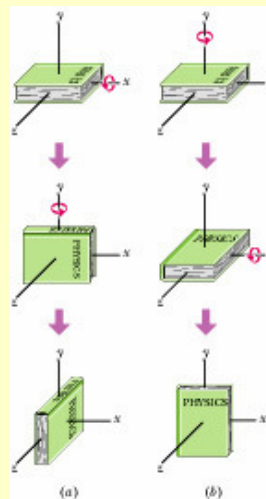
- What is the angular speed of (a) second hand, (b) minute hand, and (c) hour hand of a smoothly running analogue watch? Give your answer in rad/s

$$\omega = \frac{\Delta\theta}{\Delta t}$$

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Are Angular Quantities Vectors?

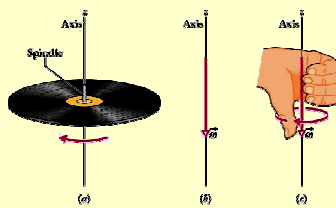
- Position, velocity and acceleration of a single particle – described by vectors. +ve and –ve signs are used to indicate direction.
- Rigid bodies can rotate about a fixed axis – either clockwise or counter-clockwise (+ or – signs are used again).
- Angular velocity ω and the angular acceleration α can be treated as vectors, but angular displacement is not a vector quantity.



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Are Angular Quantities Vectors?

- **RH Rule:** Curl the fingers of RH in direction of rotation. Your thumb points along the direction of the angular velocity vector.
- Read section 10-3 for more detail.



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Constant Angular Velocity

- $\omega = \frac{d\theta}{dt}$
 $d\theta = \omega dt$
 $\int_{\theta_0}^{\theta} d\theta = \omega \int_{t=0}^t dt$
 $[\theta]_{\theta_0}^{\theta} = \omega [t]_0^t$
- $\theta - \theta_0 = \omega t$
- For 1 revolution, $t = T$ and $\theta = 2\pi$
 $\therefore 2\pi = 0 + \omega T$
 $T = \frac{2\pi}{\omega} = \text{period}$
 $f = \frac{1}{T} = \frac{\omega}{2\pi} = \text{frequency}$

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Constant Angular Acceleration

- Derivation of these equations for this special case is the same as those for linear motion with constant acceleration.

Translational Motion Rotational Motion

$$x \leftrightarrow \theta$$

$$v \leftrightarrow \omega$$

$$a \leftrightarrow \alpha$$

$$v = v_0 + at \leftrightarrow \omega = \omega_0 + \alpha t \quad (\text{eq. 1})$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2 \leftrightarrow \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \quad (\text{eq. 2})$$

$$v^2 - v_0^2 = 2a(x - x_0) \leftrightarrow \omega^2 - \omega_0^2 = 2\alpha(\theta - \theta_0) \quad (\text{eq. 3})$$

$$x = x_0 + \frac{1}{2}(v + v_0)t \leftrightarrow \theta = \theta_0 + \frac{1}{2}(\omega + \omega_0)t \quad (\text{eq. 4})$$

$$x = x_0 + vt - \frac{1}{2}at^2 \leftrightarrow \theta = \theta_0 + \omega t - \frac{1}{2}\alpha t^2 \quad (\text{eq. 5})$$

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Linear and Angular Variables.

Position: $s = \theta r$

Speed: $\frac{ds}{dt} = \frac{d\theta}{dt} r$
 $v = \omega r$

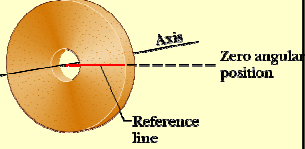
Period: $\frac{2\pi r}{v} = \frac{2\pi}{\omega}$

Acceleration: $\frac{dv}{dt} = \frac{d\omega}{dt} r$
 $a_t = \alpha r$
 $a_r = \frac{v^2}{r} = \omega^2 r$

Radians!!

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Sample Problem p248

- A grindstone rotates at constant angular acceleration $\alpha = 0.35 \text{ rad/s}^2$. At time $t = 0$, it has an angular velocity of $\omega_0 = -4.6 \text{ rad/s}$ and a reference line on it is horizontal, at the angular position $\theta_0 = 0$.

(a) At what time after $t = 0$ is the reference line at the angular position $\theta = 5 \text{ rev}$?

(b) Describe the grindstone's rotation between $t = 0$ and $t = 32 \text{ s}$.

(c) At what time t does the grindstone momentarily stop?

 - $\alpha = 0.35 \text{ rad/s}^2$
 - $\omega_0 = -4.6 \text{ rad/s}$
 - $\theta_0 = 0$

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Problem 6 (8th Ed values)

- The angular position of a point on a rotating wheel is given by $\theta = 2.0 + 4.0t^2 + 2.0t^3$, where θ is in radians and t is in seconds. At $t = 0$, what are (a) the points angular position and (b) its angular velocity? (c) What is its angular velocity at $t = 4.0 \text{ s}$? (d) Calculate its angular acceleration at $t = 2.0$. (e) Is its angular acceleration constant?

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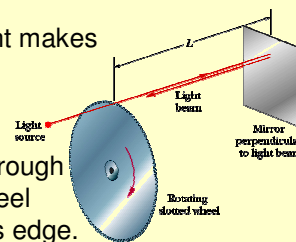
Problem 13

- A flywheel turns through 40 rev as it slows from an angular speed of 1.5 rad/s to a stop.
 - (a) Assuming a constant angular acceleration, find the time for it to come to rest.
 - (b) What is its angular acceleration?
 - (c) How much time is required for it to complete the first 20 of the 40 revolutions?
 - $\Delta\theta = 40 \text{ rev} = 80 \pi$
 - $\omega_0 = 1.5 \text{ rad/s}$

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Problem 29

- An early method of measuring the speed of light makes use of a rotating slotted wheel. A beam of light passes through one of the slots at the outside edge of the wheel, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.0 cm, and 500 slots around its edge. Measurements taken when the mirror is $L = 500\text{m}$ from the wheel indicated a speed of light $3 \times 10^5 \text{ km/s}$ (a) What is the constant angular speed of the wheel? (b) What is the linear speed of a point on the edge of the wheel?



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Rotational KE

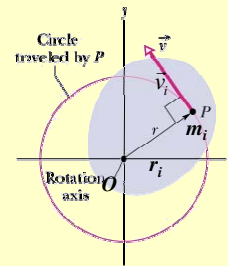
- Consider the rotating rigid body
- Divide body into small parts of masses $m_1, m_2, m_3, \dots m_i$
- Kinetic energy of rotation

$$K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + \dots + \frac{1}{2}m_iv_i^2$$

$$= \sum \frac{1}{2}m_iv_i^2 = \sum \frac{1}{2}m_i(\omega r_i)^2 = \frac{1}{2}(\sum m_ir_i^2)\omega^2$$

- Mass** - resistance of object to **change in velocity**
- Rotational Inertia** (moment of Inertia) – resistance of an object to changes in ω : $I = \sum m_ir_i^2$
- Rotational KE:

$$K = \frac{1}{2}I\omega^2$$



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Rotational Inertia – TABLE 10-2

Hoop about central axis $I = MR^2$ (a)	Annular cylinder (or ring) about central axis $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	Solid cylinder (or disk) about central axis $I = \frac{1}{2}MR^2$ (c)
Solid cylinder (or disk) about central diameter $I = \frac{1}{2}MR^2 + \frac{1}{12}ML^2$ (d)	Thin rod about axis through center perpendicular to length $I = \frac{1}{12}ML^2$ (e)	Solid sphere about any diameter $I = \frac{2}{5}MR^2$ (f)
Thin spherical shell about any diameter $I = \frac{2}{3}MR^2$ (g)	Hoop about any diameter $I = \frac{1}{2}MR^2$ (h)	Slab about perpendicular axis through center $I = \frac{1}{12}M(a^2 + b^2)$ (i)

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Rotational Inertia

- For a continuous mass:

$$\rho = \frac{dm}{dV} \quad \therefore dm = \rho dV$$

- Then

$$\begin{aligned} I &= \int R^2 dm \\ &= \int \rho R^2 dV && \text{-- 3-dimensions} \\ &= \int \sigma r^2 dV && \text{-- 2-dimensions} \\ &= \int \lambda x^2 dx && \text{-- 1-dimension} \end{aligned}$$

- ρ – Volume density
- σ – Area density
- λ – Line density

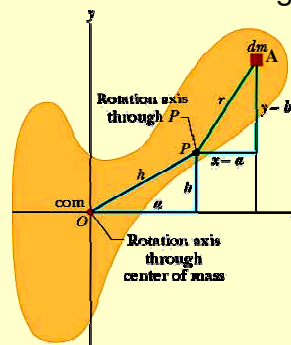
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Parallel Axis Theorem

- To find the Rotational inertia I of a body of mass M about an axis which does not pass through the centre of mass.
- If I_{com} is known then: $I = I_{com} + Mh^2$

where h is the distance from the axis to the axis through the centre of mass.

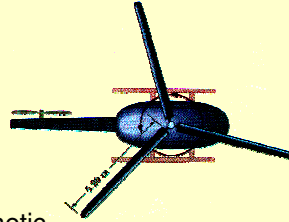
- The **moment of inertia** of a body **about any axis** is equal to the **moment of inertia** ($= Mh^2$) it would have about that axis if all the its mass were concentrated at its center of mass **plus its moment of inertia** ($= I_{com}$) about a **parallel axis** through its center of mass.



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Problem

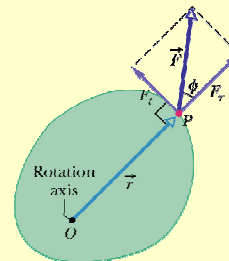
- Each of the 3 helicopter rotor blades is 5.20 m long and has a mass of 240 kg. The rotor is rotating at 350 rev/min. (a) What is the rotational inertia of the rotor assembly about the axis of rotation? (Each blade can be considered to be a thin rod.) (b) What is the kinetic energy of rotation?



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Torque

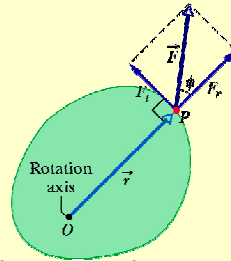
- Opening a heavy door
 - Where is door knob positioned?
 - Where do you normally apply the force?
 - What direction do you apply the force?
 - If force is not applied at the edge of the door and at 90° to the door it is more difficult to move the door.



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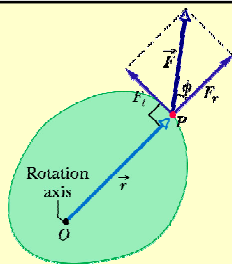
Torque

- Body is free to rotate about axis through O .
- A force $\vec{F}$ is applied at point P , positioned from O with vector $\vec{r}$.
- The vectors $\vec{F}$ and $\vec{r}$ make an angle ϕ with each other.
- How does $\vec{F}$ result in rotation of the body.
- Resolve $\vec{F}$ into two components
 - radial component - no rotation $F_r = F \cos \phi$
 - tangential component - rotation $F_t = F \sin \phi$
- The ability of force to rotate body depends on magnitude of force and where it is applied
- Torque is defined as: $\tau = (r)(F \sin \phi) = (r \sin \phi)(F)$



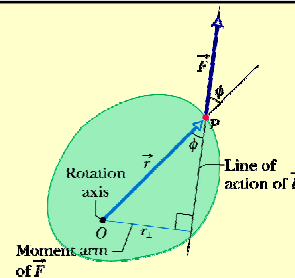
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Torque



$$\tau = (r)(F \sin \phi) = rF_t$$

$$\tau = (r \sin \phi)(F) = r_{\perp} F$$



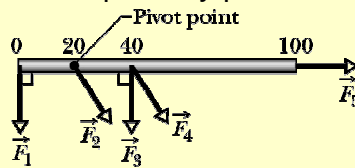
- $r_{\perp} \equiv$ moment arm
- Torque is the magnitude of the force multiplied by the perpendicular distance from the axis of rotation to the line of action of the force.
- Torque can be written as a vector cross product:

$$\vec{\tau} = \vec{r} \times \vec{F}$$
- **Right hand rule** - move $\vec{F}$ to O , curl fingers from $\vec{r}$ to $\vec{F}$, thumb shown direction of $\vec{\tau}$.

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Checkpoint 6

- The Figure shows an overhead view of a meter stick that can pivot about the dot at the position marked 20 (for 20 cm). All five forces on the stick are horizontal and have the same magnitude. Rank the forces according to the magnitude of the torque they produce, greatest first.



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Newton's Second Law for Rotation

- A force acts on a particle, which moves in circular path, therefore only the tangential component can accelerate the particle.

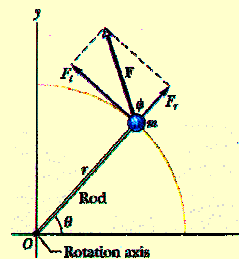
- $F_t = ma_t$

- $\tau = F_t r = ma_t r$

- $\tau = m(\alpha r)r = (mr^2)\alpha$

- $\tau = I \alpha$

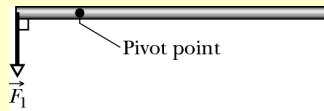
$$\sum \tau = I \alpha$$



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Checkpoint 7

- The figure shows an overhead view of a meter stick that can pivot about the point indicated which is to the left of the stick's midpoint. Two horizontal forces are applied to the stick. Only one is shown. The second force is perpendicular to the stick and is applied to the right end. If the stick is ont to turn (a) what should the direction of the force be and (b) should this second force be greater or less than or equal to the first?



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Sample Problem p261

- The figure shows a uniform disk with mass $M = 2.5 \text{ kg}$ and radius $R = 20 \text{ cm}$, mounted on a fixed horizontal axle. A block with mass $m = 1.2 \text{ kg}$ hangs from a massless cord that is wrapped around the rim of the disk. Find the acceleration of the falling block, the angular acceleration of the disk and the tension in the cord. The cord does not slip, and there is no friction at the axle.



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Work and Rotational Kinetic Energy

- $\Delta K = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2 = W$
- $W = \int_{\theta_i}^{\theta_f} \tau d\theta$
- $P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \omega$

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Problem

- Attached to each end of a thin steel rod of length 1.20m and mass 6.40 kg is a small ball of mass 1.06 kg. The rod is constrained to rotate in a horizontal plane about a vertical axis through its midpoint. At a certain instant, it is observed to be rotating with an angular velocity of 39.0 rev/s. Because of friction, it comes to rest 32s later. Assuming constant frictional torque, compute (a) the angular acceleration, (b) the retarding torque exerted by the friction, (c) the total mechanical energy dissipated by the friction, and (d) the number of revs executed during the 32.0 s. (e) Now suppose that the frictional torque is known not to be constant, which, if any, of the quantities above can still be computed without additional information?

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Problem

- (a) α , (b) τ exerted by the friction, (c) the total mechanical energy dissipated by the friction, and (d) the number of revs executed during the 32.0 s. (e) Now suppose that the frictional torque is known not to be constant, which, if any, of the quantities above can still be computed without additional information?